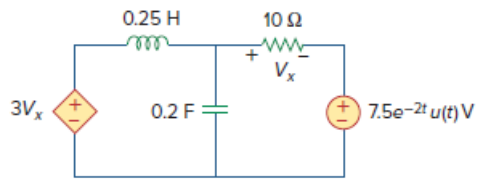


Problem

Find $V_X(s)$ in the circuit shown in Fig. 16.71.

Figure 16.71:



Step-by-step solution

Step 1 of 4

(Refer Figure 16.71 from the textbook) Transform the circuit in s-domain.

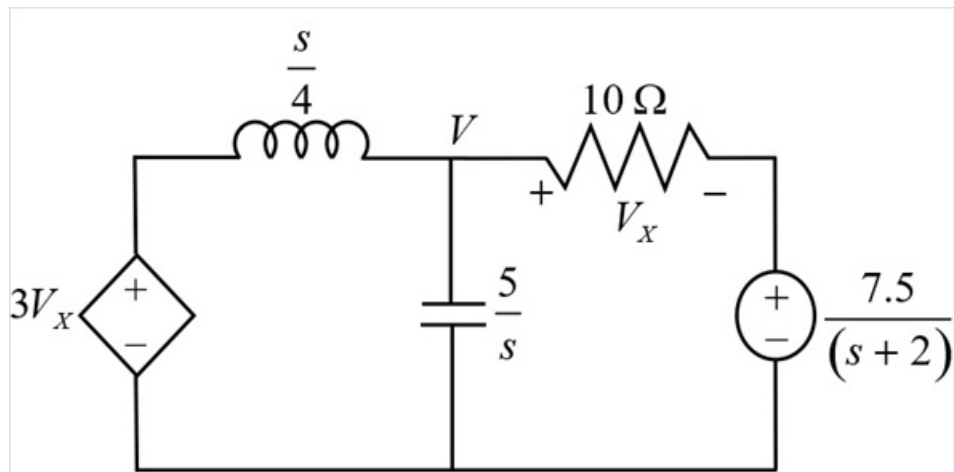


Figure 1

Step 2 of 4

Apply the nodal analysis in the Figure 1.

$$\frac{V - 3V_x}{\left(\frac{s}{4}\right)} + \frac{V}{\left(\frac{5}{s}\right)} + \frac{V - \left(\frac{7.5}{s+2}\right)}{10} = 0$$

$$\frac{4V - 12V_x}{s} + \frac{sV}{5} + \frac{V - \left(\frac{7.5}{s+2}\right)}{10} = 0$$

$$\frac{40V - 120V_x + 2s^2V + s\left[V - \left(\frac{7.5}{s+2}\right)\right]}{10s} = 0$$

$$40V - 120V_x + 2s^2V + sV - \left(\frac{7.5s}{s+2}\right) = 0$$

Rewrite the above equation.

$$V(2s^2 + s + 40) - 120V_x - \left(\frac{7.5s}{s+2}\right) = 0 \dots\dots (1)$$

Step 3 of 4

From Figure 1, the node voltage V is,

$$V = V_x + \frac{7.5}{(s+2)}$$

Substitute $V_x + \frac{7.5}{(s+2)}$ for V in equation (1).

$$\left(V_x + \frac{7.5}{(s+2)}\right)(2s^2 + s + 40) - 120V_x - \left(\frac{7.5s}{s+2}\right) = 0$$

$$V_x(2s^2 + s + 40 - 120) + \frac{15s^2 + 7.5s + 300 - 7.5s}{(s+2)} = 0$$

$$V_x(2s^2 + s - 80) = \frac{-15s^2 - 300}{(s+2)}$$

$$V_x = \frac{-(15s^2 + 300)}{(s+2)(2s^2 + s - 80)}$$

Step 4 of 4

Rewrite the above equation.

$$\begin{aligned} V_x &= \frac{-15(s^2 + 20)}{2(s+2)(s^2 + 0.5s - 40)} \\ &= \frac{-7.5(s^2 + 20)}{(s+2)(s^2 + 0.5s - 40)} \end{aligned}$$

Hence, the voltage $V_x(s)$ is $\boxed{\frac{-7.5(s^2 + 20)}{(s+2)(s^2 + 0.5s - 40)} \text{ V}}$.